SUMS OF TWO SQUARES WITH FEW PRIME FACTORS, II: THE PETERSSON NORM OF THE CUSPIDAL PART OF $\theta^2(mz)\theta^2(m'z)$, THE LEVEL OF DISTRIBUTION 1/2, AND ALMOST ALL N

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ABSTRACT. Let m, m' be coprime odd integers, $\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z)$, $G_{m,m'}$ its cuspidal part and $L = 4mm'$, so that $a_G(N) = T(m, m'; N) - M(m, m'; N)$ is the error term in the number of representations of N by $m(x^2 + y^2) + m'(z^2 + w^2)$. We prove $\|G_{m,m'}\| \ll_\varepsilon (mm')^\varepsilon$ for the Petersson norm. The proof combines Waibel's decomposition of the norm over the cusps of $\Gamma_0(L)$ with the observation that at every cusp the expansion of $\Theta_{m,m'}$ is again a theta series of a form $A(x^2 + y^2) + B(z^2 + w^2)$ with $AB \asymp mm'$, whose coefficients are $O(n^\varepsilon)$ in the range that matters. Waibel's general bound gives $(mm')^{1/4+\varepsilon}$ for this family, and numerically $\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2$. By the Hecke-operator argument of Part I this proves the conjecture of Part I: $T(m, m'; N) - M(m, m'; N) \ll d(N)N^{1/2}(mm')^\varepsilon$ uniformly, so that the weighted sequence $r(n)r(N - n)$ has level of distribution $1/2$, and the main theorem of Part I holds with $\Omega(n(N - n)) \le 8$ and $\max(\Omega(n), \Omega(N - n)) \le 5$ for every large $N \equiv 2 \pmod 4$. We also prove a mean-square bound $\sum_{N \le Y} |a_G(N)|^2 \le 4\pi e^2 Y(1 + 54Y/L)\|G\|^2$ by a covering argument, which gives the level of distribution $1 - \varepsilon$ for all but $O(X^{1-\delta})$ integers $N \le X$, and for those N the bounds 5 and 4.

1. RESULTS

We keep the notation of [1], referred to as Part I. Thus $r(n)$ is the number of representations of n as a sum of two squares, m, m' are coprime odd positive integers,

$$T(m, m'; N) = \sum_{ma+m'b=N} r(a)r(b), \qquad L = 4mm',$$

$$\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z) = \sum_{N \ge 0} T(m, m'; N)e(Nz) \in M_2(\Gamma_0(L)),$$

$\Theta_{m,m'} = E + G$ is the decomposition into the Eisenstein part, whose coefficients are the Siegel main terms $M(m, m'; N)$, and the cuspidal part $G = G_{m,m'} \in S_2(\Gamma_0(L))$, so that

$$a_G(N) = T(m, m'; N) - M(m, m'; N) \qquad \text{for every } N \ge 1.$$

The Petersson norm is $\|G\|^2 = \int_{\Gamma_0(L)\backslash\mathbb{H}} |G|^2 \,\mathrm{d}x\,\mathrm{d}y$. Lemma 2.1 of Part I, which is Waibel's theorem [3, Theorem 1] for the form $Q_{m,m'} = m(x^2+y^2)+m'(z^2+w^2)$, gives $\|G\| \ll (mm')^{1/4+\varepsilon}$, and Part I deduces from it, through a Hecke operator, the uniform bound $a_G(N) \ll d(N)N^{1/2}(mm')^{1/4+\varepsilon}$ for $(N, mm') = 1$, $v_2(N) \le 1$, and the level of distribution $2/5$. Part I conjectures (Conjecture 5.1 there) that the factor $(mm')^{1/4}$ can be removed. The first result of this paper is that the norm itself is much smaller than Waibel's general bound.

Theorem 1.1 (The norm)**.** *For coprime odd positive integers m, m',*

$$\|G_{m,m'}\|^2 \ \ll_\varepsilon \ (mm')^\varepsilon.$$

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The proof (Section 2) uses Waibel's decomposition of the norm over the cusps of $\Gamma_0(L)$ and his description of the Fourier expansion of a theta series at a cusp [3, (22) and Lemma 12]. For $Q_{m,m'}$ the expansion at every cusp is again a theta series of a form $A(x_1^2+x_2^2)+B(x_3^2+x_4^2)$ with $AB \asymp mm'$, and such a form has only $O(n^{\varepsilon})$ representations of every $n \ll mm'$, because the equation $A\alpha+B\beta=n$ has $O(1)$ solutions (α,β). This is special to our family; in general the Fourier coefficients at a cusp are of size $n^{1+\varepsilon}/\sqrt{\det}$, which is what Waibel's bound reflects. Numerically, $\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2$ (Section 7).

Corollary 1.2 (Conjecture 5.1 of Part I, level 1/2, and the bounds 8 and 5)**.** (i) *For coprime odd m,m' and $N\ge 1$ with $(N,mm')=1$ and $v_2(N)\le 1$,*

$$T(m,m';N) = M(m,m';N) + O_{\varepsilon}\big(d(N)\,N^{1/2}(mm')^{\varepsilon}\big).$$

This is Conjecture 5.1 of Part I. For coprime odd cube-free m,m' and every N with $v_2(N)\le 1$ the error term is $O_{\varepsilon}(d(N)^4N^{1/2}(mm')^{\varepsilon})$.

(ii) *The weighted sequence $r(n)r(N-n)$ has level of distribution $1/2$: Theorem 3.2 of Part I holds with $D=N^{1/2-\varepsilon}$.*

(iii) *Theorem 1.1 of Part I holds with 9 and 6 replaced by 8 and 5: for every sufficiently large $N\equiv 2 \pmod 4$,*

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 8,\ \max(\Omega(n),\Omega(N-n))\le 5\big\} \ge\ 0.23\,\mathfrak{S}_r(N)\,\frac{N}{(\log N)^2}, \qquad \mathfrak{S}_r(N)=\prod_{\substack{p\mid N\\ p>2}}\frac{p-1}{p-2},$$

the integers counted being coprime to N and products of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{6/125}$.

Part (i) is Theorem 1.1 inserted into Lemma 2.2 of Part I; (ii) is the proof of Theorem 3.2 of Part I with the exponent $1/4$ replaced by ε; (iii) is Remark 4.5 of Part I, whose sieve parameters at the level $1/2$ were certified there in interval arithmetic. The details are in Section 3.

The second theme of the paper is the average over N of the error term, which is of independent interest and gives a stronger level of distribution for almost all N.

Theorem 1.3 (Mean square)**.** *Let $L\ge 1$, $G\in S_2(\Gamma_0(L))$ and $Y\ge 1$. Then*

$$\sum_{n\le Y}|a_G(n)|^2 \ \le\ 4\pi e^2\,Y\Big(1+\frac{54\,Y}{L}\Big)\|G\|^2. \tag{1}$$

In particular, for coprime odd m,m', $G=G_{m,m'}$ and $Y\ge mm'$, by Theorem 1.1,

$$\sum_{N\le Y}\big|T(m,m';N)-M(m,m';N)\big|^2 \ \ll_{\varepsilon}\ Y^2\,(mm')^{-1+\varepsilon}, \tag{2}$$

and hence, since $\sum_{Y/2<N\le Y}|a_G(N)|^2/N \le (2/Y)\sum_{N\le Y}|a_G(N)|^2$, the root mean square of $(T-M)/\sqrt{N}$ over $Y/2<N\le Y$ is $\ll (mm')^{-1/2+\varepsilon}$.

The proof of (1) is a classical covering argument and does not use the Petersson formula; the dependence on the level is exactly the factor Y/L, i.e. the volume of $\Gamma_0(L)\backslash\mathbb{H}$. The bound (2) is the exact order of magnitude: by the Rankin–Selberg method the mean square of $(T-M)/\sqrt{N}$ is asymptotic to $16\pi^2\|G\|^2/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$, and Table 1 shows the data.

Averaging over N is natural for the arithmetic application, since the sieve of Part I is applied to one N at a time and an exceptional set of N can be tolerated. Let, as in Part I, $N\equiv 2$

(mod 4), $\mathcal{A}=\{n\le N:\ (n,N)=1\}$, $w(n)=r(n)r(N-n)$, and for coprime square-free d,e with $(de,2N)=1$ let $R(d,e)=R(d,e;N)$ be the remainder in

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n,\ e\mid N-n}} w(n)=X_N\,h(d)h(e)+R(d,e;N)$$

(this is $S(d,e)=M(d,e)+R(d,e)$ of Part I, Section 3; we write X_N for the main term X there, to avoid a clash with the length X of the range of N below).

Theorem 1.4 (Level of distribution $1-\varepsilon$ for almost all N)**.** *Let $0<\varepsilon<1/20$, $X\ge X_0(\varepsilon)$ and $D=X^{1-3\varepsilon}$. Then*

$$\sum_{\substack{X/2<N\le X\\ N\equiv 2\ (4)}}\ \sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \ll_\varepsilon\ X^{2-\varepsilon/2}.$$

Consequently, for all but $O_\varepsilon(X^{1-\varepsilon/4})$ integers $N\equiv 2$ (mod 4) in $(X/2,X]$,

$$\sum_{\substack{de\le N^{1-3\varepsilon}\\ (d,e)=1,\ (de,2N)=1}} \mu^2(de)\,d_3(de)\,|R(d,e;N)| \ \le\ N^{1-\varepsilon/5}. \tag{3}$$

The sieve of Part I, Section 4, uses the level of distribution only through (3); running it at the level $\theta=19/20$ gives the following.

Corollary 1.5. *For all but $O(X^{1-\delta})$ integers $N\equiv 2$ (mod 4) in $(X/2,X]$, with an absolute $\delta>0$,*

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 5,\ \max(\Omega(n),\Omega(N-n))\le 4\big\} \ge\ 0.34\,\mathfrak{S}_r(N)\,\frac{N}{(\log N)^2},$$

the integers counted being coprime to N and products of distinct primes $\equiv 1$ (mod 4) exceeding $N^{2/25}$.

The certified sieve parameters are $\alpha=2/25$, $\lambda=1/3$, $\beta_1=69/100$ (Section 5). The pairs $(5,4)$ for almost all N and $(8,5)$ for every N should be compared with $(9,6)$ for every N in Part I. A floating-point scan (Figure 2) shows that $(5,4)$ is the limit of the Richert weights of Part I even at the level 1; a bound such as $(\cdot,3)$ would need different weights.

Section 6 restates the uniform bound of Corollary 1.2(i) in terms of Poincaré series, and Section 7 contains the numerical evidence: the data of Part I, an exact check of the whole numerical chain at the level 20, and the empirical law $\|G_{m,m'}\|^2\approx 0.37(\log mm')^2$, which corrects a remark of Part I.

2. Proof of Theorem 1.1

Throughout this section $Q=Q_{m,m'}$ has Gram matrix $A_Q=\operatorname{diag}(2m,2m,2m',2m')$, $\det A_Q=16(mm')^2$, level $L=4mm'$, and $G=\Theta_{m,m'}-E$. We use the following three facts from [3], in Waibel's notation ($m=4$ variables, weight $m/2=2$; his N is our L, his Q our A_Q).

Lemma 2.1 ([3, (22)])**.** *Let $\mu=[\mathrm{SL}_2(\mathbb{Z}):\Gamma_0(L)]$, $\tilde\mu=[\mathrm{SL}_2(\mathbb{Z}):\Gamma(L)]$, so that $\tilde\mu/\mu=L\varphi(L)$, let $\mathcal{F}$ be the standard fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ and $T=\{\left(\begin{smallmatrix}1&x\\0&1\end{smallmatrix}\right):\ 0\le x\le L-1\}$. For a,c modulo L with $(a,c,L)=1$ let $\varrho_{a,c}=\left(\begin{smallmatrix}a&*\\c&*\end{smallmatrix}\right)\in\mathrm{SL}_2(\mathbb{Z})$. Then for $g\in S_2(\Gamma_0(L))$*

$$\|g\|^2=\frac{\mu}{\tilde\mu}\sum_{c\ (L)}\ \sum_{\substack{a\ (L)\\ (a,c,L)=1}}\ \int_{\bigcup_{t\in T}t\mathcal{F}}\big|(g|_2\varrho_{a,c})(z)\big|^2\,\mathrm{d}x\,\mathrm{d}y.$$

This is the fundamental domain of $\Gamma(L)$ written as a union of translates of $\mathcal{F}$ by coset representatives $\varrho_{a,c}t$ of $\Gamma(L)\backslash \mathrm{SL}_2(\mathbb{Z})$, together with $\langle g,g\rangle_{\Gamma_0(L)} = [\Gamma_0(L):\Gamma(L)]^{-1}\langle g,g\rangle_{\Gamma(L)}$ and the invariance of $y^2|g|^2$.

Lemma 2.2 ([3, Lemma 12 and p. 15])**.** *Let $\varrho = \varrho_{a,c}$ with $c>0$ and $d=(c,L)<L$, so that $\varrho \notin \Gamma_0(L)$. There are $\hat{d}\in\{d,d/2\}$, a diagonal integral matrix $D=D_\varrho$ and a positive integral quaternary form $S=S_\varrho$ of level at most L, depending on c only through d and the parity data of c, such that*

$$(G|_2\varrho)(z) = \sum_{n\geq 1} b_\varrho(n)\, e\Big(\frac{\hat{d}nz}{L}\Big), \qquad |b_\varrho(n)|^2 \leq 32\,\frac{\det D}{\det A_Q}\big(r(S,n)^2 + r(\mathrm{gen}S,n)^2\big),$$

where $r(S,n)$ is the number of representations of n by S and $r(\mathrm{gen}S,n)$ the Siegel average over the genus of S. Explicitly, $S=(L/\hat{d})\,D\tilde{A}^{-1}D$, where $\tilde{A}=U^{\mathrm{T}}A_QU$ with $U\in\mathrm{SL}_4(\mathbb{Z})$ chosen so that $\frac{1}{2}x^{\mathrm{T}}\tilde{A}x \equiv \sum_i q_i x_i^2$ modulo the odd part $\tilde{c}$ of c and is in Jordan form modulo $2^{v_2(c)}$, and $D=\mathrm{diag}(\tilde{d}_i 2^{t_i})$ with $\tilde{d}_i=(q_i,\tilde{c},L)$ and $t_i\in\{0,1\}$ determined by $v_2(c)$.

Here the factor $32=2\cdot 2^m$ accounts for the weights $\eta(x)\in\{0,1,\sqrt{2}\}$ in Waibel's expansion and for $|u-v|^2\leq 2|u|^2+2|v|^2$; the Eisenstein part $E=\theta(\mathrm{gen}Q)$ transforms into the genus theta series of S, which is how $r(\mathrm{gen}S,n)$ enters ([3, p. 15]).

Lemma 2.3 ([3, (14)])**.** *For a positive integral quaternary form S of level L_S,*

$$r(\mathrm{gen}S,n) \ll_\varepsilon \frac{n\,(n,L_S)^{1/2}}{\sqrt{\det S}}\,(nL_S)^\varepsilon.$$

We now specialise Lemma 2.2 to $Q_{m,m'}$.

Lemma 2.4. *For $Q=Q_{m,m'}$ one can take $U=I$ in Lemma 2.2, and then, with $d_m=(m,c)$, $d_{m'}=(m',c)$ and $d_2=(c,4)$, so that $d=d_2d_md_{m'}$,*

$$D=\mathrm{diag}(d_m,d_m,d_{m'},d_{m'})\cdot 2^t, \quad t\in\{0,1\}, \qquad \tfrac{1}{2}x^{\mathrm{T}}Sx = A(x_1^2+x_2^2)+B(x_3^2+x_4^2),$$

where $A=\kappa\,(m'/d_{m'})\,d_m$ and $B=\kappa\,(m/d_m)\,d_{m'}$ are positive integers with $\kappa = 4^t d_m d_{m'}/\hat{d} \in \{\frac{1}{4},\frac{1}{2},1,2,4,8\}$. In particular

$$\det D \leq 16\,d^2, \qquad AB=\kappa^2 mm' \geq \frac{mm'}{16}, \qquad \frac{(A,B)}{AB}\leq\frac{4}{mm'}, \qquad \sqrt{\det S}=4AB\geq\frac{mm'}{4}.$$

Proof. The form $\frac{1}{2}x^{\mathrm{T}}A_Qx = m(x_1^2+x_2^2)+m'(x_3^2+x_4^2)$ is already diagonal, so $U=I$, $q=(m,m,m',m')$, there are no binary 2-adic blocks, and all 2-adic valuations $\nu_i(2)$ vanish; hence $\tilde{d}_i=(q_i,\tilde{c},L)=(q_i,c)$, which is d_m for $i=1,2$ and $d_{m'}$ for $i=3,4$, and the t_i are all equal to one value $t\in\{0,1\}$. Then $A_Q^{-1}=\mathrm{diag}(1/(2m),1/(2m),1/(2m'),1/(2m'))$ and

$$\tfrac{1}{2}x^{\mathrm{T}}Sx = \frac{L}{\hat{d}}\cdot 4^t\Big(\frac{d_m^2(x_1^2+x_2^2)}{4m}+\frac{d_{m'}^2(x_3^2+x_4^2)}{4m'}\Big) = \frac{4^t}{\hat{d}}\big(m'd_m^2(x_1^2+x_2^2)+md_{m'}^2(x_3^2+x_4^2)\big),$$

which is the stated form with $\kappa = 4^t d_md_{m'}/\hat{d} = 4^t/\hat{d}_2$, $\hat{d}_2=\hat{d}/(d_md_{m'})\in\{d_2,d_2/2\}\subset\{\frac{1}{2},1,2,4\}$. The integrality of S (Waibel's Lemma 11) forces $A,B\in\mathbb{Z}$. Since $(m'/d_{m'})d_m$ and $(m/d_m)d_{m'}$ are coprime, $(A,B)\leq\kappa$ if $\kappa\in\mathbb{Z}$ and $(A,B)=1$ otherwise, so $(A,B)/(AB)\leq \kappa/(\kappa^2mm') = 1/(\kappa mm') \leq 4/(mm')$. Finally $\det D = (d_md_{m'})^2 2^{4t}\leq 16d^2$ and $\det S = \det\mathrm{diag}(2A,2A,2B,2B)=16A^2B^2$. ∎

Lemma 2.5. *Let $A,B\geq 1$ be integers with $(A,B)/(AB)\leq 4/(mm')$. Then for every $n\geq 1$,*

$$r\big(A(x_1^2+x_2^2)+B(x_3^2+x_4^2),\,n\big) \ll_\varepsilon n^\varepsilon\Big(1+\frac{n}{mm'}\Big).$$

Proof. A representation gives $\alpha = x_1^2 + x_2^2$, $\beta = x_3^2 + x_4^2$ with $A\alpha + B\beta = n$, $\alpha, \beta \geq 0$. The solutions α of this equation lie in one residue class modulo $B/(A,B)$ and in $[0, n/A]$, so there are at most $n(A,B)/(AB) + 1 \leq 4n/(mm') + 1$ pairs (α, β), and each pair contributes $r(\alpha)r(\beta) \leq 16\, d(\alpha)d(\beta) \ll n^{\varepsilon}$ representations. □

Proof of Theorem 1.1. Apply Lemma 2.1 to $g = G$. Fix $\varrho = \varrho_{a,c}$ and put $I(\varrho) = \int_{\cup_t t\mathcal{F}} |G|_2\varrho|^2 \,\mathrm{d}x\,\mathrm{d}y$. The region $\cup_{t\in T} t\mathcal{F}$ lies in the strip $\{-\frac{1}{2} \leq x \leq L - \frac{1}{2},\ y \geq \sqrt{3}/2\}$, and $G|_2\varrho$ is periodic in x with a period dividing L, being a form on $\varrho^{-1}\Gamma_0(L)\varrho \supseteq \Gamma(L)$. Hence, if $G|_2\varrho = \sum_n b(n) e(\hat{d}nz/L)$, Parseval over a full period gives

$$I(\varrho) \leq L \sum_{n\geq 1} |b(n)|^2 \int_{\sqrt{3}/2}^{\infty} e^{-4\pi\hat{d}ny/L}\,\mathrm{d}y = \frac{L^2}{4\pi\hat{d}} \sum_{n\geq 1} \frac{|b(n)|^2}{n}\, e^{-2\sqrt{3}\pi\hat{d}n/L}. \tag{4}$$

The cusp ∞. If $d = L$, then $\varrho \in \Gamma_0(L)$, $b(n) = a_G(n)$ and $\hat{d} = L$. For every n, $|a_G(n)| \leq T(m,m';n) + M(m,m';n) \ll n^{\varepsilon}(1 + n/(mm'))$ by Lemma 2.5 (for T) and Lemma 2.3 applied to Q itself (for M; here $(n,L)^{1/2} \leq n^{1/2}$ and $n^{3/2}/(mm') \leq n^{3/2}$ is harmless because of the exponential). Hence $I(\varrho) \ll L\sum_n n^{2\varepsilon}(1 + n^{3/2})e^{-2\sqrt{3}\pi n} \ll L$, and these $\varphi(L)$ pairs (a,c) contribute $\ll (\mu/\tilde{\mu})\varphi(L)L = 1$ to the sum in Lemma 2.1.

The other cusps. Let $d \mid L$, $d < L$. The number of c modulo L with $(c,L) = d$ is $\varphi(L/d)$, and for each such c the number of a modulo L with $(a,c,L) = 1$ is $\varphi(d)L/d$; since $\varphi(L/d)\varphi(d) \leq \varphi(L)$, these pairs contribute at most

$$\frac{\mu}{\tilde{\mu}}\,\varphi(L/d)\,\varphi(d)\,\frac{L}{d}\ \max_{\varrho} I(\varrho)\ \leq\ \frac{1}{d}\max_{\varrho} I(\varrho),$$

the maximum over the $\varrho_{a,c}$ with $(c,L) = d$. By Lemma 2.2, Lemma 2.4 and (4), with $L^2 = 16(mm')^2$, $\det A_Q = 16(mm')^2$, $\det D \leq 16d^2$ and $d/\hat{d} \leq 2$,

$$\begin{aligned}\frac{1}{d}I(\varrho) &\leq \frac{L^2}{4\pi d\hat{d}} \cdot \frac{32\cdot 16d^2}{16(mm')^2} \sum_n \frac{r(S,n)^2 + r(\mathrm{gen}S,n)^2}{n} e^{-2\sqrt{3}\pi\hat{d}n/L} \\ &\leq \frac{256}{\pi} \sum_n \frac{r(S,n)^2 + r(\mathrm{gen}S,n)^2}{n} e^{-2\sqrt{3}\pi\hat{d}n/L}.\end{aligned}$$

For the first sum, Lemma 2.5 gives $r(S,n)^2 \ll n^{2\varepsilon}(1 + n/(mm'))^2$, and since $\hat{d} \geq d/2 \geq 1/2$ and $n/(mm') \leq 8$ for $n \leq 2L$,

$$\begin{aligned}\sum_n \frac{n^{2\varepsilon}(1+n/(mm'))^2}{n} e^{-2\sqrt{3}\pi\hat{d}n/L} &\ll \sum_{n\leq 2L} n^{2\varepsilon-1}\Big(1 + \frac{n}{mm'}\Big)^2 + \frac{L^{2\varepsilon}}{(mm')^2} \sum_{n>2L} n\, e^{-\sqrt{3}\pi n/L} \\ &\ll L^{2\varepsilon}\log L + L^{2\varepsilon}\frac{L^2}{(mm')^2} \ll L^{3\varepsilon}.\end{aligned}$$

For the second sum, Lemma 2.3 with $L_S \leq L$ and $\sqrt{\det S} \geq mm'/4$ gives $r(\mathrm{gen}S,n)^2 \ll L^{2\varepsilon}n^2(n,L)/(mm')^2$, and

$$\begin{aligned}\sum_n \frac{n^2(n,L)}{(mm')^2 n} e^{-2\sqrt{3}\pi\hat{d}n/L} &\leq \frac{1}{(mm')^2} \sum_{g\mid L} g \sum_{k\geq 1} gk\, e^{-\sqrt{3}\pi gk/L} \\ &\ll \frac{1}{(mm')^2} \sum_{g\mid L} g^2\Big(\frac{L}{g}\Big)^2 = \frac{d(L)L^2}{(mm')^2} \ll L^{\varepsilon},\end{aligned}$$

where we wrote $n = gk$ with $g = (n, L)$ and used $\sum_k k\, e^{-\lambda k} \ll \lambda^{-2}$. Hence each class $d < L$ contributes $\ll L^{3\varepsilon}$, there are $d(L) \ll L^{\varepsilon}$ classes, and

$$\|G\|^2 \ll 1 + d(L) L^{3\varepsilon} \ll L^{4\varepsilon}.$$

Renaming ε gives the theorem. □

Remark 2.6. The proof shows more precisely that

$$\|G_{m,m'}\|^2 \ll d(mm')\,(\log mm')^2 \max_{n \le 8mm'} \max_{\alpha,\beta \le n} \big(d(\alpha)d(\beta)\big)^2,$$

so that the true order of the norm is governed by the size of $r(\alpha)r(\beta)$ on the $O(1)$ pairs (α, β) with $A\alpha + B\beta = n$, $n \ll mm'$. On average over n these products are of size $(\log n)$, which suggests $\|G_{m,m'}\|^2 \asymp (\log mm')^2$; the data of Section 7 give $\|G_{m,m'}\|^2 \approx 0.37(\log mm')^2$. Waibel's Theorem 1 gives $(mm')^{1/2+\varepsilon}$ for $\|G\|^2$, and his Theorem 2 for diagonal forms gives $\min(m, m')(mm')^{\varepsilon}$; both are sharp for other families and both lose here because they bound $r(S, n)$ by the general lattice-point estimate of [3, Lemma 9].

3. Proof of Corollary 1.2

(i). Lemma 2.2 of Part I states that for $L \ge 1$, $G \in S_2(\Gamma_0(L))$ and $n = n_1 n_2$ with $(n_1, L) = 1$, $n_2 \mid L^{\infty}$, $n_2 \le L^{1/2}$, one has $|a_G(n)| \le C\, d(n_1) n^{1/2} \|G\|$ with an absolute constant C. For $G = G_{m,m'}$, $L = 4mm'$, $(N, mm') = 1$, $v_2(N) \le 1$, Part I applies it with n_1 the odd part of N and $n_2 = 2^{v_2(N)}$, and Theorem 1.1 bounds $\|G\|$. This is the first assertion. Lemma 2.5 of Part I (the exact reduction by the operators U_p) then gives the second, exactly as in the proof of Corollary 2.7 of Part I, with $(mm')^{1/4+\varepsilon}$ replaced by $(mm')^{\varepsilon}$ throughout.

(ii). In the proof of Theorem 3.2 of Part I, the exponent $1/4$ enters only through the bound (19) there, $|T - M| \ll 4^{\omega(N)} d(N)^2 N^{1/2} (\tilde m \tilde m')^{1/4+\eta}$, which by (i) now holds with $(\tilde m \tilde m')^{\eta}$. With $D = N^{1/2-\varepsilon}$ and $\eta = \varepsilon/10$ the terms with $t_1 s s' \le N^{\varepsilon}$ contribute, in the notation of that proof,

$$\ll N^{\eta} \cdot N^{\eta} \cdot N^{1/2} \cdot N^{\eta} \sum_{k \le D} d(k) k^{\eta} \ll N^{1/2+4\eta} D^{1+\eta} \ll N^{1-\varepsilon+5\eta} = N^{1-\varepsilon/2},$$

and the terms with $t_1 s s' > N^{\varepsilon}$ contribute $\ll N^{1-\varepsilon+5\eta} + N^{5\eta} D \ll N^{1-\varepsilon/2}$ as before. Hence Theorem 3.2 of Part I holds with $D = N^{1/2-\varepsilon}$ and the bound $N^{1-\varepsilon/3}$.

(iii). Proposition 4.3 of Part I and its proof use the level only through the conclusion of Theorem 3.2, with θ the exponent of D; they hold with $\theta = 1/2$ by (ii). Remark 4.5 of Part I records the parameters

$$\theta = \tfrac{1}{2}, \qquad \alpha = \tfrac{6}{125}, \qquad \lambda = \tfrac{1}{3}, \qquad \beta_1 = \tfrac{17}{50}, \qquad \tfrac{1}{\lambda} + \tfrac{2}{\beta_1} = \tfrac{151}{17} < 9, \qquad \tfrac{1}{\lambda} + \tfrac{1}{\beta_1} = \tfrac{101}{17} < 6,$$

for which `certify_modular.py` of Part I (argument `1/2`) certifies $L \ge 0.997718$, $R \le 2.498893$ and $L - \lambda R \ge 0.164754$ in interval arithmetic. By Lemma 4.2 of Part I this gives $\Omega(n(N-n)) \le 8$ and $\max(\Omega(n), \Omega(N-n)) \le 5$, the weight of each n counted is $w(n) = 16 \cdot 2^{\Omega(n(N-n))} \le 2^{12}$, and as in the proof of Theorem 1.1 of Part I the number of integers counted is at least

$$\big(0.164754 + o(1)\big) 2^{-12} \cdot 8\, \mathfrak{S}_r(N)\, c_{2h} \frac{N}{\alpha^2 (\log N)^2} \ge 0.23\, \mathfrak{S}_r(N) \frac{N}{(\log N)^2},$$

since $0.164754 \cdot 2^{-12} \cdot 8 \cdot 1.66485 \cdot (125/6)^2 = 0.2325\ldots$. □

Remark 3.1. Corollary 1.2 removes the conditional status of Remark 4.5 and of the level $1/2$ in Part I; Section 5 of Part I (the numerical evidence and the conjecture) is superseded, and its Section 5.5 (the heuristic spectral interpretation) is no longer needed for the result, although the question it raises, why the spectral sum there exhibits the cancellation, is answered by Theorem 1.1 only indirectly, through the cusps. The remark in Section 5.3 of Part I that the root

mean square of $(T-M)/\sqrt{N}$ "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp" is incorrect; see Section 7.

4. Proof of Theorem 1.3

Let $G(z)=\sum_{n\ge1}a(n)e(nz)\in S_2(\Gamma_0(L))$ and put $F(z)=y^2|G(z)|^2$, $z=x+iy$. Since $G(\gamma z)=(cz+d)^2G(z)$ and $\operatorname{Im}(\gamma z)=y/|cz+d|^2$ for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(L)$, the function F is $\Gamma_0(L)$-invariant, and with the invariant measure $\mathrm{d}\mu=y^{-2}\mathrm{d}x\,\mathrm{d}y$ we have $\|G\|^2=\int_{\mathcal F}F\,\mathrm{d}\mu$ for any fundamental domain $\mathcal F$ of $\Gamma_0(L)$.

Let $y_0=1/(4\pi Y)$ and $R=\{x+iy:\ 0\le x<1,\ y_0\le y\le 2y_0\}$ (half-open in x, so that each orbit of the translations $z\mapsto z+b$ meets R at most once). By Parseval, $\int_0^1|G(x+iy)|^2\,\mathrm{d}x=\sum_n|a(n)|^2e^{-4\pi ny}$, hence

$$\int_R|G|^2\,\mathrm{d}x\,\mathrm{d}y=\sum_n|a(n)|^2\int_{y_0}^{2y_0}e^{-4\pi ny}\,\mathrm{d}y\ \ge\ \frac{e^{-2}}{4\pi Y}\sum_{n\le Y}|a(n)|^2,$$

because $4\pi ny\le 8\pi Yy_0=2$ for $n\le Y$, $y\le 2y_0$. On the other hand $\int_R|G|^2\,\mathrm{d}x\,\mathrm{d}y=\int_R F\,\mathrm{d}\mu$, and

$$\int_R F\,\mathrm{d}\mu=\sum_{\gamma\in\Gamma_0(L)/\{\pm1\}}\int_{R\cap\gamma\mathcal F}F\,\mathrm{d}\mu=\int_{\mathcal F}F(w)\,\#\{\gamma:\ \gamma w\in R\}\,\mathrm{d}\mu(w)\ \le\ \mu_R\,\|G\|^2,$$

where $\mu_R=\sup_{z\in R}\#\{\gamma\in\Gamma_0(L)/\{\pm1\}:\ \gamma z\in R\}$ (if $\#\{\gamma:\gamma w\in R\}\ne0$ for some $w\in\mathcal F$, pick γ_0 with $z=\gamma_0 w\in R$; then $\gamma\mapsto\gamma\gamma_0^{-1}$ is a bijection from $\{\gamma:\gamma w\in R\}$ onto $\{\gamma:\gamma z\in R\}$, so the count at w is at most μ_R). It remains to bound μ_R. Fix $z=x+iy\in R$ and let $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(L)$ with $\gamma z\in R$.

- If $c=0$, then $\gamma=\pm\left(\begin{smallmatrix}1&b\\0&1\end{smallmatrix}\right)$ and $\gamma z=z+b$, so $x+b\in[0,1)$ forces $b=0$: one element.
- If $c\ne0$, we may take $c>0$, since γ and $-\gamma$ are the same element of $\Gamma_0(L)/\{\pm1\}$. Then $L\mid c$, and $\operatorname{Im}(\gamma z)=y/|cz+d|^2\in[y_0,2y_0]$ together with $y\in[y_0,2y_0]$ gives $|cz+d|^2\in[1/2,2]$. Since $|cz+d|^2=(cx+d)^2+c^2y^2$, this forces $c^2y^2\le2$, hence $c\le\sqrt{2}/y_0=4\sqrt{2}\,\pi Y$, and $|cx+d|\le\sqrt{2}$. The number of positive multiples c of L with $c\le4\sqrt{2}\pi Y$ is $\lfloor4\sqrt{2}\pi Y/L\rfloor\le4\sqrt{2}\pi\,Y/L$; for each c there are at most 3 integers d with $|cx+d|\le\sqrt{2}$; and for each pair (c,d) the elements of $\Gamma_0(L)$ with bottom row (c,d) are $\left(\begin{smallmatrix}1&k\\0&1\end{smallmatrix}\right)\gamma_0$, $k\in\mathbb Z$, whose images γ_0z+k have real part in $[0,1)$ for exactly one k. This gives at most $4\sqrt{2}\pi\cdot3\,Y/L=12\sqrt{2}\pi\,Y/L<54\,Y/L$ elements.

Hence $\mu_R\le1+54Y/L$, and

$$\sum_{n\le Y}|a(n)|^2\le4\pi e^2Y\int_R F\,\mathrm{d}\mu\le4\pi e^2\,Y\Big(1+\frac{54Y}{L}\Big)\|G\|^2,$$

which is (1). For $G=G_{m,m'}$, $L=4mm'$ and $Y\ge mm'$ the bracket is $\le1+54Y/(4mm')\le15Y/(mm')$, and Theorem 1.1 gives (2). □

Remark 4.1. The argument is the classical one for bounding $\sum_{n\le Y}|a(n)|^2$ by the norm (compare [2, Chapter 5]); its only input is the invariance of $y^2|G|^2$. The strip $y_0\le y\le2y_0$ with $y_0=1/(4\pi Y)$ is chosen so that the exponential factor in Parseval's identity stays at e^{-2} for all $n\le Y$. For $Y\ge L$ the factor Y^2/L in (1) is of the correct order of magnitude: by the Rankin–Selberg method, $\sum_{n\le Y}|a(n)|^2/n\sim16\pi^2\|G\|^2Y/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb H)$ with $\operatorname{vol}=\frac{\pi}{3}L\prod_{p\mid L}(1+1/p)$ (measure $\mathrm{d}\mu$), so that $\sum_{n\le Y}|a(n)|^2\sim8\pi^2\|G\|^2Y^2/\operatorname{vol}$ for fixed G as $Y\to\infty$. Theorem 1.3 makes this uniform in the level, at the cost of the constant.

5. Proof of Theorem 1.4 and Corollary 1.5

5.1. **The remainder on average.** We follow the proof of Theorem 3.2 of Part I, replacing the uniform bound for $|T-M|$ by Theorem 1.3 after an application of Cauchy–Schwarz over N. Recall from Part I, (15), that for $N\equiv 2 \pmod 4$ and coprime square-free d,e with $(de,2N)=1$,

$$R(d,e;N)=\sum_{t\mid N^\flat}\mu(t)\sum_{m\mid d^2}\sum_{m'\mid e^2}c_d(m)c_e(m')\sum_{\substack{t_1\mid t^2\\ t\mid t_1}}c_t(t_1)\big(a_G(N)-a_G(N/2)\big),\qquad G=G_{t_1m,m'},\tag{5}$$

where $N^\flat$ is the product of the odd primes dividing N, $m=ds$, $m'=es'$ with $s\mid d$, $s'\mid e$, and likewise $t_1=tu$ with $u\mid t$ (the coefficients $c_d(m)$, $c_t(t_1)$ of Part I, (14), vanish unless $m=ds$ with $s\mid d$, resp. $t_1=tu$ with $u\mid t$; in particular $t\mid t_1$); the parameters t_1m,m' of the form $Q_{t_1m,m'}$ are coprime, odd and cube-free, $\tilde m\tilde m':=t_1mm'=t_1ss'\cdot de$, and the coefficients c satisfy $|c_d(m)c_e(m')c_t(t_1)|\le 3^{\omega(d)+\omega(e)+\omega(t)}$. (For the reader's convenience: c_d is the multiplicative coefficient of Part I, (14), with $c_p(p)=2$, $c_p(p^2)=-1$ for a split prime p and $c_q(q)=0$, $c_q(q^2)=1$ for an inert prime q; it encodes the identities $r(n)=2r(n/p)-r(n/p^2)$ and $r(n)=r(n/q^2)$, which turn the condition $d\mid n$ into a combination of theta series $\theta^2(mz)$. The sum over t is the Möbius inversion of the condition $(n,N)=1$, and $a_G(N)-a_G(N/2)$ selects the odd n.) Let $\eta=\varepsilon/10$. For $N\le X$ with $X\ge X_0(\varepsilon)$ we have, as in (F1)–(F3) of Part I: $d_3(de)\,3^{\omega(d)+\omega(e)+\omega(t)}\,4^{\omega(N)}d(N)^2\le X^\eta$; for fixed (d,e,N) the number of tuples (s,s',t,t_1) is at most X^η; and $\sum_{k\le D}d(k)/k\le X^\eta$, $\sum_{k\le D}d(k)k^{-1/2}\le D^{1/2}X^\eta$. We also use the two bounds for a single term,

$$|a_G(N')|\le T(\tilde m,\tilde m';N')+M(\tilde m,\tilde m';N')\ll X^{2\eta}\Big(\frac{X}{\tilde m\tilde m'}+1\Big)\qquad (N'\le X),\tag{6}$$

$$\sum_{n\le X}|a_G(n)|\le X^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}\ll X\Big(1+\frac{X}{L}\Big)^{1/2}L^\eta,\tag{7}$$

where $G=G_{\tilde m,\tilde m'}$ and $L=4\tilde m\tilde m'$, the first being (20) of Part I, the second Cauchy–Schwarz, (1) and Theorem 1.1; for $L\le X$ the right-hand side of (7) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/2+\eta}$. Let Σ denote the double sum in Theorem 1.4. Inserting (5) and splitting the tuples according to the size of t_1ss':

Tuples with $t_1ss'>X^\varepsilon$. For each N and each tuple, (6) with $\tilde m\tilde m'>X^\varepsilon de$ gives $|a_G(N')|\ll X^{2\eta}(X^{1-\varepsilon}/(de)+1)$. Summing over the $\le X^\eta$ tuples, over $N\le X$ and over $de\le D$ with the weights $d_3(de)\cdot 3^\omega\le X^\eta$, these tuples contribute

$$\ll X^{1+4\eta}\sum_{k\le D}d(k)\Big(\frac{X^{1-\varepsilon}}{k}+1\Big)\ll X^{2-\varepsilon+5\eta}+X^{1+5\eta}D\ll X^{2-\varepsilon/2}.$$

Tuples with $t_1ss'\le X^\varepsilon$. Now $t\le t_1\le X^\varepsilon$ (as $t\mid t_1$), so t runs over at most X^ε values, t_1 over at most $d(t^2)\le X^\eta$ values, and (s,s') over at most $d(d)d(e)\le X^\eta$ values; the forms $G=G_{t_1m,m'}$ have level $L=4t_1ss'de\le 4X^\varepsilon D\le X$, so that the right-hand side of (7) is $\ll X^{3/2}(\tilde m\tilde m')^{-1/2+\eta}$. For a fixed tuple and fixed (d,e), we drop the condition $t\mid N$ except for the count: the number of $N\le X$ with $t\mid N$ is $\le X/t$, and Cauchy–Schwarz gives

$$\begin{aligned}\sum_{\substack{N\le X\\ t\mid N}}\big(|a_G(N)|+|a_G(N/2)|\big)&\le 2\Big(\frac{X}{t}\Big)^{1/2}\Big(\sum_{n\le X}|a_G(n)|^2\Big)^{1/2}\\ &\ll\Big(\frac{X}{t}\Big)^{1/2}X\,(\tilde m\tilde m')^{-1/2+\eta}\le X^{3/2+\eta}\,t^{-1/2}(de)^{-1/2},\end{aligned}$$

using (1) in the form $\sum_{n\le X}|a_G(n)|^2 \ll X^2\|G\|^2/L$ and $(\tilde m\tilde m')^{-1/2+\eta}\le X^\eta(de)^{-1/2}$. Summing over $t\le X^\varepsilon$ ($\sum_t t^{-1/2}\le 2X^{\varepsilon/2}$), over the other tuple variables ($\le X^{2\eta}$), and over $de\le D$ with the weights $\le X^\eta$,

$$\ll X^{3/2+4\eta+\varepsilon/2}\sum_{k\le D} d(k)k^{-1/2} \ll X^{3/2+5\eta+\varepsilon/2}D^{1/2} = X^{2-\varepsilon+5\eta} = X^{2-\varepsilon/2},$$

since $D^{1/2}=X^{1/2-3\varepsilon/2}$. Both contributions are $\ll X^{2-\varepsilon/2}$, which proves the first assertion. (The margin is small but positive: the choice $D=X^{1-3\varepsilon}$ saves $X^{-3\varepsilon/2}$ in $D^{1/2}$, of which the sum over t costs $X^{\varepsilon/2}$ and the divisor-type factors cost $X^{5\eta}=X^{\varepsilon/2}$; this is why $\eta=\varepsilon/10$.)

For the second, let $S(N)$ be the inner sum with $D=X^{1-3\varepsilon}$. The number of $N\in(X/2,X]$, $N\equiv 2 \pmod 4$, with $S(N)>X^{1-\varepsilon/4}$ is at most $\Sigma/X^{1-\varepsilon/4}\ll X^{1-\varepsilon/4}$. For the remaining N, since $N^{1-3\varepsilon}\le D$ and $X^{1-\varepsilon/4}\le(2N)^{1-\varepsilon/4}\le N^{1-\varepsilon/5}$ for large X, (3) holds. □

Remark 5.1. The exponent 1 is the balance $X^{3/2}D^{1/2}=X^2$ between the mean-square bound $X^{3/2}(de)^{-1/2}$ for the sum over N of one remainder and the number D of moduli; with a norm bound $\|G_{m,m'}\|\ll(mm')^{\beta+\varepsilon}$ in place of Theorem 1.1 one gets the level $1/(2(1+\beta))$ for every N (Part I, Remark 3.3) and $1/(1+2\beta)$ for almost all N; Waibel's $\beta=1/4$ gives $2/5$ and $2/3$, and $\beta=0$ gives $1/2$ and 1. The exceptional set $X^{1-\varepsilon/4}$ comes from Markov's inequality applied to a first moment of $S(N)$, which in turn rests on the second moment of $a_G(N)$; a bound for a higher moment would shrink it, and we do not pursue this.

5.2. **The sieve at the level** 19/20**.** Proposition 4.3 of Part I and its proof use the level of distribution only through the bound (3) (there with 2/5); they hold verbatim, for a given N, with $\theta=2/5$ replaced by any $\theta\in(0,1)$ for which (3) holds with $N^{\theta-\varepsilon}$ in place of $N^{1-3\varepsilon}$, provided $\alpha<\beta_1<\theta$. Lemmas 4.1 and 4.2 of Part I do not involve the level at all. Hence, for every N satisfying (3) with $3\varepsilon<1/20$, the conclusion of Proposition 4.3 holds with $\theta=19/20$, and the proof of Theorem 1.1 of Part I goes through with the parameters certified for $\theta=19/20$. The script `certify_modular.py` of Part I, extended by the argument `19/20` in the repository of this paper, certifies in interval arithmetic

$$\theta=\tfrac{19}{20},\qquad \alpha=\tfrac{2}{25},\qquad \lambda=\tfrac{1}{3},\qquad \beta_1=\tfrac{69}{100},$$
$$\tfrac{1}{\lambda}+\tfrac{2}{\beta_1}=\tfrac{407}{69}<6,\qquad \tfrac{1}{\lambda}+\tfrac{1}{\beta_1}=\tfrac{307}{69}<5,\qquad L-\lambda R\ge 0.085058$$

(with $L\ge 0.999484$, $R\le 2.743278$, $\theta_A=0.4740$; the output files are listed at the end of the paper). By Lemma 4.2 of Part I this gives $\Omega(n(N-n))\le 5$ and $\max(\Omega(n),\Omega(N-n))\le 4$, and the weight of each n counted is $w(n)=16\cdot 2^{\Omega(n(N-n))}\le 2^9$. As in the proof of Theorem 1.1 of Part I, the number of integers counted is at least

$$\big(0.085058+o(1)\big)\,2^{-9}\cdot 8\,\mathfrak{S}_r(N)\,c_{2h}\,\frac{N}{\alpha^2(\log N)^2}\ \ge\ 0.34\,\mathfrak{S}_r(N)\frac{N}{(\log N)^2},$$

since $0.085058\cdot 2^{-9}\cdot 8\cdot 1.66485\cdot(25/2)^2=0.3457\ldots$. The exceptional set is that of Theorem 1.4 with, say, $\varepsilon=1/80$, so $\delta=1/320$ is admissible. □

Remark 5.2. The same sieve gives $(8,5)$ at the level 1/2 (Corollary 1.2), $(6,4)$ at 2/3 (certified: $\alpha=3/50$, $\lambda=1/3$, $\beta_1=51/100$, $L-\lambda R\ge 0.036741$) and $(5,4)$ at 19/20. The floating-point scan of Figure 2 shows that the best attainable value of $1/\lambda+1/\beta_1$ stays above 4 for every $\theta<1$, so $\max\Omega\le 3$ is out of reach of the Richert weights with one parameter β_1, and that $1/\lambda+2/\beta_1$ drops below 6 only for $\theta>0.83$. Corollary 1.5 is an "almost all N" statement with a power-saving exceptional set; it says nothing about any individual N.

6. The uniform bound and the Poincaré series

Let $L = 4mm'$ and let $P_N \in S_2(\Gamma_0(L))$, $N \geq 1$, be the Riesz representative of the N-th coefficient: the unique cusp form with

$$\langle f, P_N \rangle = \frac{a_f(N)}{4\pi N} \qquad \text{for all } f \in S_2(\Gamma_0(L)), \tag{8}$$

i.e. $P_N = (4\pi N)^{-1} \sum_{f \in \mathcal{B}} \overline{a_f(N)}\, f$ for an orthonormal basis $\mathcal{B}$. (This is the N-th Poincaré series; for weight 2 the series itself converges only conditionally, which is why we define P_N by (8).) By (11) of Part I,

$$\|P_N\|^2 = \frac{a_{P_N}(N)}{4\pi N} = \frac{1}{(4\pi N)^2} \sum_{f \in \mathcal{B}} |a_f(N)|^2 = \frac{\Delta_L(N,N)}{4\pi N},$$

where $\Delta_L(N,N)$ is the Petersson density. Hence, for $G = G_{m,m'}$,

$$|a_G(N)|^2 = (4\pi N)^2 |\langle G, P_N \rangle|^2 = 4\pi N\, \Delta_L(N,N)\, \|G\|^2 \cos^2 \angle(G, P_N), \tag{9}$$

where $\cos \angle(G, P_N) = |\langle G, P_N \rangle| / (\|G\|\, \|P_N\|)$. The identity separates the three quantities that enter any bound for $a_G(N)$: the norm $\|G\|^2$, which by Theorem 1.1 is $\ll L^{\varepsilon}$; the Petersson density, which is $\leq C'$ for $N \leq L^{1/2}$ (Part I, Lemma 2.2, Step 4) and $1 + O(N^{1/2+\varepsilon}/L)$ in general; and the alignment $\cos^2 \angle(G, P_N) \leq 1$. Corollary 1.2(i) is obtained from (9) by bounding the alignment trivially: after the coprime part of N is moved into a Hecke operator (Part I, Lemma 2.2), only P_{n_2} with $n_2 \in \{1, 2\}$ occurs, where the density is bounded, and the norm does the rest. Theorem 1.3 says what the alignment is on average: inserting (9) into (1) and restricting to $N \in (Y/2, Y]$,

$$\frac{1}{Y/2} \sum_{Y/2 < N \leq Y} \Delta_L(N,N) \cos^2 \angle(G, P_N) \;\leq\; 4e^2 \Big(\frac{1}{Y} + \frac{54}{L} \Big) \;\ll\; \frac{1}{L} \qquad (Y \geq L), \tag{10}$$

the value of $\cos^2$ for a unit vector in general position in the $\asymp L$-dimensional space $S_2(\Gamma_0(L))$: on average over N, the theta series is no more aligned with the Poincaré series than a random cusp form of level L would be, and Corollary 1.2(i) says that no single N is aligned by more than $N^{\varepsilon} L / \|G\|^2 \gg L^{1-\varepsilon}$ times the average. Expanding in an orthogonal basis $\mathcal{B}^*$ of Hecke eigenforms as in Part I, Section 5.5,

$$\langle G, P_N \rangle = \sum_{\varphi \in \mathcal{B}^*} \frac{\langle G, \varphi \rangle \langle \varphi, P_N \rangle}{\|\varphi\|^2} = \frac{1}{4\pi N} \sum_{\varphi \in \mathcal{B}^*} \frac{\langle \Theta, \varphi \rangle\, a_\varphi(N)}{\|\varphi\|^2},$$

so that Corollary 1.2(i) is the bound $\ll N^{1/2+\varepsilon}$ for this spectral sum, uniformly in m, m'; Part I asked for a proof through the cancellation between the coefficients $\langle \Theta, \varphi \rangle$ and $a_\varphi(N)$. The proof given here does not see this cancellation: it shows instead that $\sum_\varphi |\langle \Theta, \varphi \rangle|^2 / \|\varphi\|^2 = \|G\|^2$ is itself small, i.e. that the theta series has a small cuspidal projection to begin with, and the spectral sum is then bounded by Cauchy–Schwarz.

7. Numerical evidence

7.1. **The mean square and the norm.** The script `mean_square_check.py` computes, with the tables and the FFT convolution of `sup_error_scan.py` of Part I, the sums $S_2(Y) = \sum_{N \in \mathcal{N}_Y} |a_G(N)|^2$ over the admissible set

$$\mathcal{N}_Y = \{ N \leq Y : \ (N, mm') = 1,\ v_2(N) \leq 3 \},$$

on which the main term of that script is exact, for $Y = 2^{21}$ and $Y = 2^{20}$, and the estimate

$$\|G\|_{\mathrm{est}}^2 = \frac{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})}{16\pi^2} \cdot \frac{1}{\#\mathcal{N}_Y} \sum_{N\in\mathcal{N}_Y} \frac{|a_G(N)|^2}{N}$$

suggested by the Rankin–Selberg asymptotic of Remark 4.1. Replacing $1/Y$ by $1/\#\mathcal{N}_Y$ assumes that the mean square of $|a_G(N)|^2/N$ over the excluded N (those with $(N, mm') > 1$ or $v_2(N) \geq 4$) is the same as over $\mathcal{N}_Y$; this is immaterial when mm' has only large prime factors, but introduces a bias of unknown sign for $(1,5)$, $(9,1)$, $(3,5)$ and $(1,15015)$, where 25%, 38%, 50% and 64% of the N are excluded. Table 1 shows the result at $Y = 2^{21}$, and Figure 1 the dependence on Y and on mm'.

(m,m')	mm'	$\#\mathcal{N}_Y$	$S_2(Y)/S_2(Y/2)$	$\|G\|_{\mathrm{est}}^2$	$\|G\|_{\mathrm{est}}^2/(mm')^{1/2}$	$\|G\|_{\mathrm{est}}^2/(\log mm')^2$
$(1,5)$	5	1572864	4.001	0.74	0.33	0.29
$(9,1)$	9	1310720	4.001	1.47	0.49	0.30
$(3,5)$	15	1048576	4.001	3.44	0.89	0.47
$(1,101)$	101	1946614	4.000	7.95	0.79	0.373
$(13,29)$	377	1752262	4.001	13.09	0.67	0.372
$(1,1009)$	1009	1964131	3.976	17.96	0.57	0.375
$(61,67)$	4087	1904986	3.961	24.47	0.38	0.354
$(101,103)$	10403	1927715	3.948	29.81	0.29	0.348
$(1,15015)$	15015	754220	4.359	36.62	0.30	0.396
$(1009,1013)$	1022117	1962192	4.559	46.87	0.046	0.244

TABLE 1. Mean square of the cuspidal error over the admissible $N \leq Y = 2^{21}$ (`results/mean_square_check_x21.txt`). The ratio $S_2(Y)/S_2(Y/2) \approx 4$ is the growth Y^2 of Theorem 1.3. For the first three pairs and for $(1,15015)$ a large proportion of the N (25% to 64%) is excluded by the coprimality condition, and for the last pair $Y < L$, so that neither (1) in its Y^2/L form nor the Rankin–Selberg asymptotic applies; these rows are shown for completeness. The exact value for $(1,5)$ is 0.5785 (Section 7.2).

7.2. **An exact check at the level** 20**.** The space $S_2(\Gamma_0(20))$ is one-dimensional, spanned by the newform $f_{20} = \eta(2z)^2\eta(10z)^2$ of the elliptic curve $20a1$: $y^2 = x^3 + x^2 + 4x + 4$, which is $X_0(20)$ itself (modular degree 1, Manin constant 1). Hence $G_{1,5} = c\, f_{20}$ with $c = a_G(1) = T(1,5;1) - M(1,5;1) = 4 - \frac{4}{3} = \frac{8}{3}$ (the Siegel product at $n = 1$ is $\frac{\pi^2}{5} \cdot \frac{25}{3\pi^2} \cdot \frac{4}{5}$), and $\|G_{1,5}\|^2 = \frac{64}{9}\|f_{20}\|^2$. The script `norm_level20.py` computes $\|f_{20}\|^2$ in two independent ways: from the Rankin–Selberg asymptotic $\sum_{n\leq y} a_f(n)^2/n \sim 16\pi^2\|f\|^2 y/\,\mathrm{vol}$, $\mathrm{vol}(\Gamma_0(20)\backslash\mathbb{H}) = 12\pi$, with the coefficients of the eta product up to 2^{20}, which gives 0.08128, 0.08128, 0.08134 at $y = 2^{16}, 2^{18}, 2^{20}$; and from the period lattice Λ of $20a1$, since $\varphi^*\omega = 2\pi i f(z)\mathrm{d}z$ gives $\|f\|^2 = \mathrm{covol}(\Lambda)/(4\pi^2)$ for a modular parametrisation of degree 1. The periods $\omega_1 = 2.824375\ldots$ (real) and $\omega_2 = 2.274165\ldots i$ of $y^2 = (x+1)(x^2+4)$ are computed by quadrature, the lattice $\mathbb{Z}\omega_1 + \mathbb{Z}(\omega_1+\omega_2)/2$ is identified as the period lattice by recomputing $g_2 = -44/3$ and $|g_3| = 296/27$ from it through E_4 and E_6, and its covolume $3.2115478\ldots$ gives $\|f_{20}\|^2 = 0.0813495\ldots$. The two values agree to four decimals. The script also checks that $T(1,5;N) - M(1,5;N)$ as computed by the pipeline of Part I equals $\frac{8}{3}a_f(N)$ on every admissible $N \leq 2^{20}$, to 10^{-9}. This validates the Siegel main term, the FFT computation, the Rankin–Selberg normalisation and the volume used in the estimate $\|G\|_{\mathrm{est}}^2$. The exact value $\|G_{1,5}\|^2 = 0.5785$ is below the estimate 0.74 of Table 1, as expected: the excluded N are either multiples of 5, where $|a_f(5n)| = |a_f(n)|$, or divisible by 16, where $a_f = 0$, so the admissible mean is biased upwards.

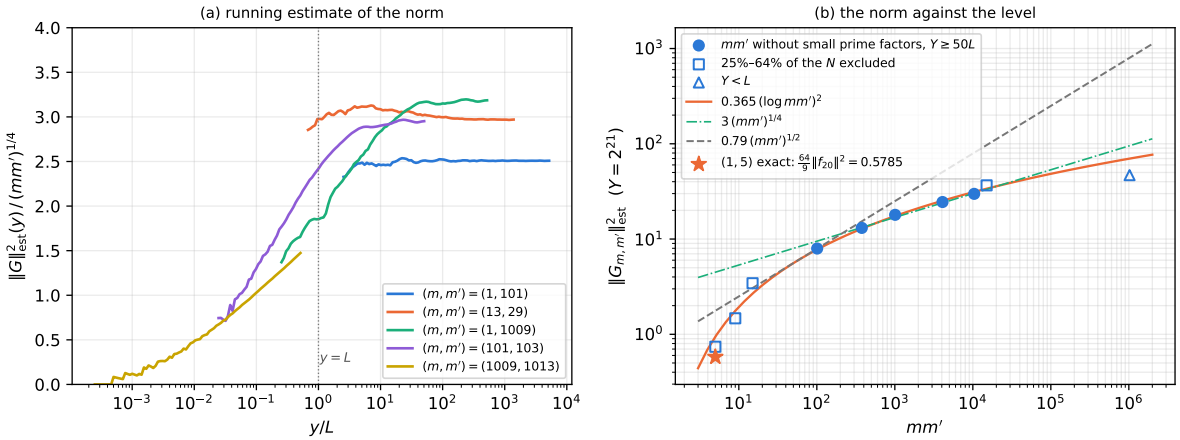


FIGURE 1. (a) The running estimate $\|G\|^2_{\mathrm{est}}(y)$, computed over the admissible $N \leq y$, divided by $(mm')^{1/4}$, against y/L on a logarithmic scale, for five pairs (m, m'); the plateau for $y \gg L$ is the growth Y^2 of Theorem 1.3 and the Rankin–Selberg asymptotic of Remark 4.1. (b) $\|G_{m,m'}\|^2_{\mathrm{est}}$ at $Y = 2^{21}$ against mm', both axes logarithmic; filled circles: the five pairs with mm' free of small prime factors and $Y \geq 50L$; open squares: pairs with 25%–64% of the N excluded; triangle: $(1009, 1013)$, for which $Y < L$; star: the exact value at $(1, 5)$. The lines are $0.365(\log mm')^2$, the fit of Section 7.3, $3(mm')^{1/4}$, and $0.79(mm')^{1/2}$ (the growth allowed by Waibel's general bound, through the point $(1, 101)$). Script `plot_mean_square.py`; floating point.

7.3. **The empirical law.** The last column of Table 1 shows $\|G\|^2_{\mathrm{est}}/(\log mm')^2$. For the five pairs with $10^2 \leq mm' \leq 1.1 \cdot 10^4$, $Y \geq 50L$ and mm' free of small prime factors, it lies between 0.348 and 0.375, with mean 0.365, while the ratio to $(mm')^{1/2}$ falls from 0.79 to 0.29 and the ratio to $(mm')^{1/4}$ rises from 2.5 to 3.2 over the same range. Over a range of a factor 100 in mm' the fits $(\log mm')^2$ and $(mm')^{1/4}$ are not separated by the data alone, but Theorem 1.1 excludes any power, and the heuristic of Remark 2.6, that $\|G\|^2$ is a sum over the cusps of $\sum_{n \ll mm'} |b(n)|^2/n$ with $|b(n)| \approx r(\alpha)r(\beta)$ on $O(1)$ pairs, predicts $(\log mm')^2$ with a constant of the order of $\frac{1}{4\pi} \cdot 2 \cdot$ (number of cusp classes with a large contribution) ≈ 0.3–0.5. We therefore record

$$\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2 \qquad (10^2 \leq mm' \leq 1.6 \cdot 10^4),$$

and, for the root mean square of $(T - M)/\sqrt{N}$ over $N \leq Y$, $Y \gg L$,

$$\mathrm{rms}^2 \approx \frac{16\pi^2 \|G\|^2}{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})} \approx \frac{44\,(\log mm')^2}{mm' \prod_{p \mid 2mm'}(1 + 1/p)},$$

which reproduces the column "rms" of Table 3 of Part I (0.27 for $(101, 103)$, 0.67 for $(1, 1009)$). This corrects the remark in Section 5.3 of Part I that the root mean square "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp": the decrease is like $(mm')^{-1/2}$ up to a logarithm, as Theorem 1.3 with Theorem 1.1 shows.

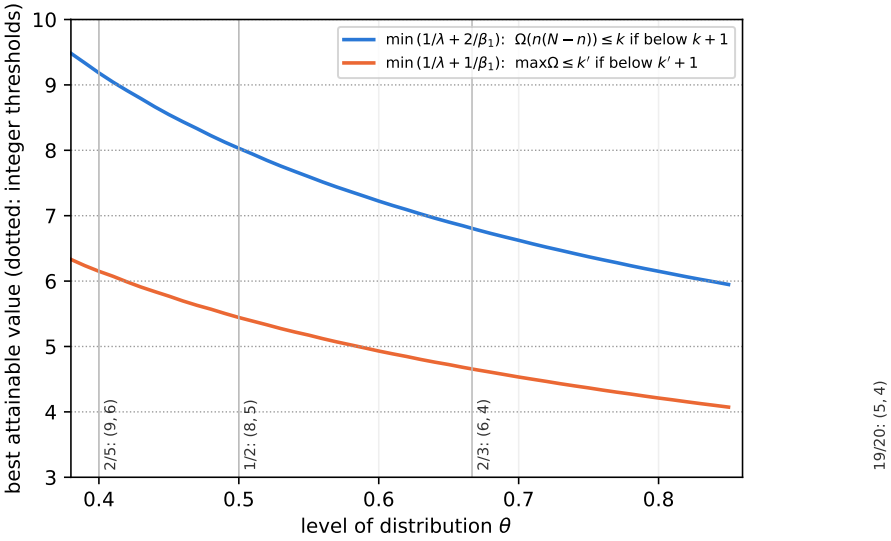


FIGURE 2. The smallest values of $1/\lambda + 2/\beta_1$ and of $1/\lambda + 1/\beta_1$ attainable with $L - \lambda R > 0$ in the sieve of Part I, as functions of the level of distribution θ (floating-point model of `modular_sieve_scan.py`, minimised over α, β_1 with $\lambda = L/R$; script `plot_sieve_thresholds.py`). The bound $\Omega(n(N-n)) \le k$, resp. $\max \Omega \le k'$, holds when the blue, resp. orange, curve is below the dotted line $k+1$, resp. $k'+1$. The vertical lines mark $\theta = 2/5$ (Part I), $1/2$ (Corollary 1.2, every N), $2/3$ and $19/20$ (almost all N); the parameter sets at these four levels are certified.

Data and code

The scripts `mean_square_check.py`, `norm_level20.py`, `certify_modular.py` (arguments `1/2`, `2/3`, `19/20`), `plot_mean_square.py` and `plot_sieve_thresholds.py` (Figures 1 and 2), the modules of Part I they use (`sup_error_scan.py`, `modular_sieve_scan.py`, `sieve_core.py`) and the outputs `results/mean_square_check_x21.txt`, `results/norm_level20.txt`, `results/certify_modular_output_twothirds.txt`, `results/certify_modular_output_nineteen.txt`, `results/sieve_thresholds.txt` are in the repository https://github.com/Ruqing1963/two-squares-mean-square; the code and data of Part I are at https://github.com/Ruqing1963/two-squares-theta-level. The certification of the sieve parameters uses directed rounding as described in Part I, Appendix A; the computations of Section 7 are in floating point and are evidence, not proof.

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